

SURFACE WATER HYDROLOGY**MODEL QUESTION PAPER**

[Time: 3 hours]

(Maximum marks: 100)

PART—A

(Maximum marks: 10)

Marks

I. Answer the following questions in one or two sentences.

- a) Name any four types of rain gauges.
- b) Define hydrology
- c) What are the classification of streams ?
- d) Name the tree forms in water is stored in a catchment
- e) Define unit hydrograph

(5 x 2=10)

PART—B

(Maximum marks: 30)

(Answer any five questions)

II a) With the aid of a sketch explain siphon rain gauge

- b) Explain evapotranspiration
- c) Explain the method of velocity measurement by floats
- d) What are the limitations of unit hydrograph? Explain
- e) Explain flow duration curve
- f) What are the various stages of river? Explain
- g) What are the classification of rivers based on variation of discharge?

Explain

(5 x 6=30)

PART—C

(Maximum marks: 60)

(Answer one full question from each module)

MODULE-I

- III. a) Explain hydrological cycle with a sketch 8
b) What are the forms of precipitation? Explain 7

OR

- IV. a) What are the weather systems associated with precipitation? Explain. 8
b) Explain standard non recording rain gauge with a sketch 7

MODULE-II

- V. a) What are the two types of current meters for measuring velocity? Explain 8
b) Discuss various factors affecting the runoff of a basin 7

OR

- VI. a) What are the methods of estimating runoff of a catchment 7
b) How do you determine the yield from a catchment and arrive at the capacity of a tank 8

MODULE-III

- VII. a) What are the steps to be adopted to derive a unit hydrograph from an observed flood hydrograph 15

OR

- VIII. a) What are the hydraulic methods for flood routing? Explain 8
b) Explain the categories of flood forecasting techniques 7

7

MODULE-IV

- IX. a) Explain the hydrological characteristics of rivers 8
b) What is river sedimentation? Explain 7

OR

- X. a) Explain the concept of watershed hydrology 8
b) What is the water quality characteristic? Explain 7